



Air moisture to water

A bike mount that can convert the air moisture into water using solar power based on thermoelectric cooling <http://dgit.in/FontusWater>

Nanoparticle Sensors

Nanoparticle sensors have been developed by a team of MIT researchers that could monitor tumours or other diseases. <http://dgit.in/NanoMonitor>

From the labs

down the US state of Virginia's James River while being escorted by 13 small guard boats. When the overhead helicopter crew detected a potential 'threat' in the form of a suspicious boat, the drone boats detected the threat via infrared sensors and quickly and efficiently manoeuvred to encircle themselves close enough to the threat so as to engage it and/or close off any potential escape or unrestricted access to the HVU.

The greatest advantage of such technology is that such an operation, which would normally require 40 people to successfully pull off, now requires just one operator. It goes without saying that the possible applications for swarm robotics are limited only by our imagination. Take, for example, the idea of nanobot swarms. What was once just a concept that could be simulated via software is now in the realm of reality. Researchers from Harvard University have conceptualised the 'Kilobot', a swarm of thousand nanobots that perform synchronised actions to rearrange themselves into various shapes. The icing



"On my way to my interview with Amazon!"

on this robotics cake is the cost efficiency of such systems. Each robot in the Kilobot costs just \$14 (₹865) and consists of two vibrating motors on four frail legs.

However, the most interesting and ground-breaking innovation in the field of swarm robotics would have to be the flying swarm drones developed by University of Pennsylvania professor Vijay Kumar and his team. Kumar's drones are lightweight, inexpensive to build and easy to program. He hopes that they can be used in emer-

gency response work by surveying buildings after an earthquake or other disaster. Although the technology has tremendous room for improvement, it's a marvellous and pioneering piece of robotic innovation.

In fact, watching these flying swarm bots co-ordinate in series after series of complex flying manoeuvres – from choreographed swarm tactics to avoiding aerial obstacles – is a truly memorable experience, one that stirs up feelings of awe and pride in our technological advances.

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